

8. (a) Let r and H be the radius and height of the cone respectively.

$$\therefore \text{Original volume} = \frac{1}{3} \pi r^2 h = V$$

New radius = r and new height = $2h$

$$\text{New volume} = \frac{1}{3} \pi r^2 \times 2h = 2V$$

$$\therefore \text{Increase in volume} = 2V - V = V$$

$$\begin{aligned} \text{Increase percentage of volume} \\ = \left(\frac{V}{V} \times 100 \right) \% = 100\% \end{aligned}$$

9. (c) Ratio of surface area

$$= \frac{4\pi r^2}{4\pi R^2} = \left(\frac{4}{25} \right) = \left(\frac{2}{5} \right)^2 \Rightarrow \frac{r}{R} = \frac{2}{5}$$

Now, ratio of volumes

$$\begin{aligned} \frac{\frac{4}{3} \pi r^3}{\frac{4}{3} \pi R^3} &= \left(\frac{r}{R} \right)^3 = \left(\frac{2}{5} \right)^3 = \frac{8}{125} \\ &= 8 : 125 \end{aligned}$$

10. (c) Let radius of cylinder = x and radius of cone = $2x$

Height of each = h

$$\begin{aligned} \therefore \text{Required ratio} &= \frac{\text{Volume of cone}}{\text{Volume of cylinder}} \\ &= \frac{\frac{1}{3} \pi 4x^2 h}{\pi x^2 h} = \frac{4}{3} = 4 : 3 \end{aligned}$$

11. (c) Volume of sphere = Volume of cylinder

$$\therefore \frac{4}{3} \pi R^3 = \pi \times 3 \times 3 \times 32$$

$$\Rightarrow R^3 = 3 \times 3 \times 3 \times 8 \Rightarrow R = 6 \text{ cm}$$

12. (c) Volume of first sphere = $\frac{4}{3} \pi \left(\frac{3}{2} \right)^3$

$$\text{and volume of second sphere} = \frac{4}{3} \pi (2)^3$$

$\therefore$ Volume of big sphere

$$= \frac{4}{3} \pi \left\{ \left(\frac{3}{2} \right)^3 + (2)^3 \right\} [\text{by condition}]$$

$$\Rightarrow \frac{4}{3} \pi x^3 = \frac{4}{3} \pi \left\{ \frac{27}{8} + 8 \right\}$$

$$x^3 = 91/8 = 11.375 \text{ cm}^3$$

13. (b) Given, $l = R$

$\therefore$ Total surface area of cone,

$$S = \pi R(R + l) = \pi R(R + R) = 2\pi R^2$$

$$\therefore \text{Area of circle, } A = \pi (2R)^2 = 4\pi R^2$$

$$\Rightarrow A = 2S$$

14. (a) In one round, distance covered by cylinder = 60 cm

In one second, distance covered by cylinder = $60 \times 5 = 300$ cm

In five seconds, distance covered by cylinder = $300 \times 5 = 1500$ cm = 15 m

15. (b) Let the radius of big drop and small drop be R and r respectively.

By given condition,

$27 \times \text{Volume of smaller drops} = \text{Volume of bigger drop}$

$$\therefore 27 \times \frac{4}{3} \pi r^3 = \frac{4}{3} \pi R^3$$

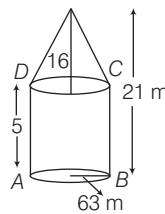
$$\Rightarrow 27 \times (0.2)^3 = R^3 \quad [\because r = 0.2 \text{ cm}]$$

$$\Rightarrow (3 \times 0.2)^3 = R^3 \Rightarrow R = 0.6 \text{ cm}$$

16. (c) Common radius = $\frac{126}{2} = 63$ m

Curved surface area of cylindrical part

$$= 2\pi rh$$



$$= 2 \times \frac{22}{7} \times 63 \times 5 = 1980 \text{ m}^2$$

Height of cone = $21 - 5 = 16$ m

Curved surface area of conical part = πrl

$$= \frac{22}{7} \times 63 \times \sqrt{(63)^2 + (16)^2}$$

$$= \frac{22}{7} \times 63 \times \sqrt{3969 + 256}$$

$$= \frac{22}{7} \times 63 \times 65 = 12870 \text{ m}^2$$

$\therefore$ Total surface area

$$= 1980 + 12870 = 14850 \text{ m}^2$$

$\therefore$ Total cost of canvas used

$$= 14850 \times 12 = ₹ 178200$$

17. (d) Given, radius of cylinder = 6 cm

Volume of sphere

$$= \frac{4}{3} \pi r^3 = \frac{4}{3} \times \frac{22}{7} \times 6 \times 6 \times 6$$

$$= \frac{88 \times 2 \times 36}{7} \text{ cm}^3$$

Let r be its internal radius, R be its external radius, then material used to cast the cylinder

$$= \pi h (R^2 - r^2) = \frac{22}{7} \times 32 (25 - r^2)$$

$$\text{Hence, } \frac{22}{7} \times 32 \times (25 - r^2)$$

$$= \frac{88 \times 2 \times 36}{7}$$

$$\Rightarrow (25 - r^2) = \frac{88 \times 2 \times 36 \times 7}{22 \times 32 \times 7} = 9$$

$$\Rightarrow r^2 = 25 - 9 = 16 \Rightarrow r = 4 \text{ cm}$$

Thickness of cylinder

$$= R - r = 5 - 4 = 1 \text{ cm}$$

18. (c) Volume of original cylinder

$$= [\pi \times (10)^2 \times 1000] \text{ cm}^3$$

Volume of shell

$$= \frac{1}{4} \times \pi \times 100000 = 25000 \pi \text{ cm}^3$$

Let r be the radius of shell, then

$$\pi r^2 h = 25000 \pi$$

$$\Rightarrow r^2 = \frac{25000 \pi}{\pi \times 1000} = 25$$

$$\Rightarrow r = 5 \text{ cm}$$

Now, thickness = $2r = 2 \times 5 = 10$ cm

$$\Rightarrow R^2 = 100 + 25 = 125$$

$$\Rightarrow R = \sqrt{125}$$

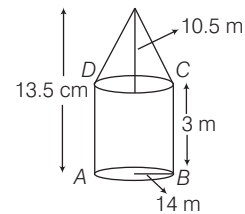
$\therefore$ Thickness of cylindrical shell

$$= R - 10 = \sqrt{125} - 10$$

$$= 5\sqrt{5} - 5 \times 2 = 5(\sqrt{5} - 2) \text{ cm}$$

19. (b) Curved surface area of cylinder

$$= 2 \times \frac{22}{7} \times 14 \times 3 = 264 \text{ m}^2$$



Height of cone = $13.5 - 3 = 10.5$ m

Slant height of cone

$$= \sqrt{(10.5)^2 + (14)^2} = \sqrt{110.25 + 196}$$

$$= \sqrt{306.25} = 17.5 \text{ m}$$

Curved surface area of cone = πrl

$$= \frac{22}{7} \times 14 \times 17.5 = 770 \text{ m}^2$$

Total area to be painted

$$= 264 + 770 = 1034 \text{ m}^2$$

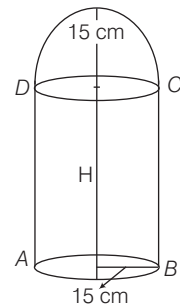
$$\therefore \text{Cost of painting} = 1034 \times 2 = ₹ 2068$$

20. (a) Let height of the cylinder be H .

By given condition,

Volume of hemisphere + Volume of cylinder = Volume of container

$$\therefore \frac{2}{3} \pi r^3 + \pi r^2 H = 32400 \pi$$



$$\Rightarrow \frac{2}{3} \pi \times 3375 + \pi \times 225 H = 32400 \pi$$

$$\Rightarrow 2\pi \times 1125 + \pi \times 225 H = 32400 \pi$$

On dividing both sides by 225π , we get

$$\Rightarrow 10 + H = 144 \Rightarrow H = 134 \text{ cm}$$

$\therefore$ Height of container

$$= 15 + 134 = 149 \text{ cm}$$

- 21. (d)** Since, milk is in a conical flask whose base radius and height are r and h , respectively.

$$\therefore \text{Volume of milk} = \frac{1}{3} \pi r^2 h$$

Now, this milk is poured into a cylindrical flask whose base radius is $2r$. Let H be the height of cylindrical flask.

$\therefore$ Volume of milk

= Volume of cylindrical flask

$$\Rightarrow \frac{1}{3} \pi r^2 h = \pi (2r)^2 H$$

$$\Rightarrow \frac{1}{3} \pi r^2 h = 4\pi r^2 H \Rightarrow H = \frac{h}{12}$$

- 22. (d)** Let radius and height of a cone be r and h .

$$\therefore \frac{r}{h} = \frac{3}{4} \Rightarrow h = \frac{4r}{3} \quad \dots(i)$$

$$\text{Volume of cone} = \frac{1}{3} \pi r^2 h$$

$$\therefore 96\pi = \frac{1}{3} \pi \times r^2 \times \frac{4r}{3}$$

$$\Rightarrow r^3 = \frac{96 \times 3 \times 3}{4} = 216 \quad [\text{from Eq. (i)}]$$

$$\Rightarrow r = 6 \text{ cm and } h = 8 \text{ cm}$$

$\therefore$ Lateral surface area = πrl

$$= \pi r \sqrt{r^2 + h^2} \quad [\because l^2 = r^2 + h^2]$$

$$= \pi \times 6 \sqrt{36 + 64} = 60\pi \text{ cm}^2$$

- 23. (b)** $\therefore$ Volume of wax balls

$$= \frac{4}{3} \pi (1)^3 = \frac{4}{3} \pi \text{ cm}^3$$

$$\text{and volume of sphere} = \frac{4}{3} \pi (8)^3 \text{ cm}^3$$

$$\therefore \text{Required number of balls} = \frac{\frac{4}{3} \pi (8)^3}{\frac{4}{3} \pi} = 512$$

- 24. (b)** Given, $r = \frac{70}{2} \text{ cm} = 0.35 \text{ m}$

and $h = 4 \text{ m}$

$\therefore$ Surface area of 10 cylinders

$$= 10 (2\pi rh)$$

$$= 10 \left(2 \times \frac{22}{7} \times 0.35 \times 4 \right) = 88 \text{ m}^2$$

$\therefore$ Total cost of 10 pillars painting at the rate of ₹ 5 per m^2

$$= 88 \times 5 = ₹ 440$$

- 25. (c)** Given,

$$R = 35 \text{ cm}, r = 14 \text{ cm and } h = 40 \text{ cm}$$

$\therefore$ Volume of the bucket

$$= \frac{\pi h}{3} (R^2 + r^2 + Rr)$$

$$= \frac{22 \times 40}{7 \times 3} (35^2 + 14^2 + 35 \times 14)$$

$$= \frac{880}{21} (1225 + 196 + 490)$$

$$= \frac{880}{21} \times 1911 = 80080 \text{ cm}^3$$

- 26. (b)** Let side of a cube be a unit.

$\therefore$ Volume of cube, $V = a^3$

and total surface area of cube, $S = 6(a)^2$

$$S^3 = 6^3 (a^6) = 216 V^2$$

- 27. (b)** Total area of wetted surface

= Curved surface area of cylinder + Area of base of cylinder

$$= 2\pi rh + \pi r^2$$

$$= \pi [2 \times 3.5 \times 4 + (3.5)^2]$$

$$= \frac{22}{7} (28 + 12.25) = \frac{22}{7} \times 40.25$$

$$= 126.5 \text{ m}^2$$

- 28. (b)** Let height h_1 , radius r_1 , area S_1 and volume V_1 of first cylinder.

Similarly, height h_2 , radius r_2 , area S_2 and volume V_2 of second cylinder.

$$\text{By given condition, } \frac{r_1}{r_2} = \frac{2}{3} \quad \dots(i)$$

$$\therefore \frac{S_1}{S_2} = \frac{5}{3} \Rightarrow \frac{2\pi r_1 h_1}{2\pi r_2 h_2} = \frac{5}{3}$$

$$\Rightarrow \frac{h_1}{h_2} \times \frac{2}{3} = \frac{5}{3} \Rightarrow \frac{h_1}{h_2} = \frac{5}{2} \quad \dots(ii)$$

$$\therefore \frac{V_1}{V_2} = \frac{\pi r_1^2 h_1}{\pi r_2^2 h_2} = \left(\frac{r_1}{r_2} \right)^2 \left(\frac{h_1}{h_2} \right) = \left(\frac{2}{3} \right)^2 \left(\frac{5}{2} \right) = \frac{10}{9} \text{ or, } 10 : 9$$

- 29. (b)** By given condition,

Volume of cylindrical vessel = Volume of sphere

$$\therefore \pi r^2 h = \frac{4}{3} \pi R^3$$

$$\Rightarrow (60)^2 \times 10 = \frac{4}{3} \left(\frac{d}{2} \right)^3$$

$$\Rightarrow (60)^2 \times 10 = \frac{4}{3} \times \frac{d^3}{8}$$

$$\Rightarrow d^3 = (60)^2 \times (60)$$

$$\therefore d = 60 \text{ cm}$$

- 30. (c)** Given, inner radius of hollow sphere

$$(r) = \frac{4}{2} = 2 \text{ cm}$$

Outer radius of hollow sphere

$$(R) = \frac{8}{2} = 4 \text{ cm}$$

Volume of metal of the sphere

$$= \frac{4}{3} \pi R^3 - \frac{4}{3} \pi r^3 = \frac{4}{3} \pi (4^3 - 2^3)$$

$$= \frac{4}{3} \pi \times 56 \text{ cm}^3$$

$$\text{Radius of base of cone } (x) = \frac{8}{2} = 4 \text{ cm}$$

$$\therefore \frac{1}{3} \pi x^2 h = \frac{4}{3} \pi \times 56 \quad [\text{by condition}]$$

$$\Rightarrow h = \frac{\frac{4}{3} \times 56 \times 3}{16} = 14 \text{ cm}$$

Hence, the height of cone is 14 cm.

- 31. (c)** Surface area of sphere, $S_1 = 4\pi r^2$

If radius is $2r$, then surface area of sphere,

$$S_2 = 4\pi (2r)^2 = 16\pi r^2$$

$$\therefore S_2 = 4S_1$$

Hence, it increases four times.

- 32. (c)** Given, $l = 6 \text{ m}$, $b = 4 \text{ m}$

and $h = 1.25 \text{ m}$

Area of wetted surface

$$= 2(l \times b + b \times h) + l \times b$$

$$= 2(7.5 \times 5) + 24$$

$$= 25 + 24 = 49 \text{ m}^2$$

- 33. (b)** Given, $R = 3 \text{ cm}$, $r = 2 \text{ cm}$, $h = 10 \text{ cm}$

Total surface area

$$= 2\pi (R + r)(R + h - r)$$

$$= 2\pi (3 + 2)(3 + 10 - 2)$$

$$= 110\pi \text{ cm}^2$$

- 34. (b)** For conical part, $r = \frac{6}{2} = 3 \text{ cm}$,

$$h = 4 \text{ cm}, l = \sqrt{h^2 + r^2} = 5 \text{ cm}$$

Surface area of conical part = πrl

$$= 3.14 \times 3 \times 5 = 47.1 \text{ cm}^2$$

For hemispherical part, $r = \frac{6}{2} = 3 \text{ cm}$

Surface area of hemispherical part

$$= 2\pi r^2 = 2 \times 3.14 \times 3 \times 3$$

$$= 56.52 \text{ cm}^2$$

$$\therefore \text{Surface area of toy} = 47.1 + 56.52 = 103.62 \text{ cm}^2$$

- 35. (b)** Let r and h be the radius and height of the cylinder. Here, $h = 9 \text{ m}$

By given condition,

$$2\pi r \times h = \frac{2\pi r}{3} (h + r)$$

$$\Rightarrow 9 = \frac{1}{3} (9 + r) \Rightarrow 27 = 9 + r$$

$$\therefore r = 18 \text{ m}$$

36. (b) Let r_1 and r_2 be the radii of two spheres.

$$\therefore \frac{\frac{4}{3}\pi r_1^3}{\frac{4}{3}\pi r_2^3} = \frac{8}{1} \Rightarrow \frac{r_1}{r_2} = \frac{2}{1}$$

$$\therefore \text{Ratio of their surface areas} = \frac{4\pi r_1^2}{4\pi r_2^2} = \left(\frac{2}{1}\right)^2 = 4:1$$

37. (d) Area of the field = Length $\times$ Breadth
 $= 12 \times 10 = 120 \text{ m}^2$

$$\text{Area of the pit's surface} = 5 \times 4 = 20 \text{ m}^2$$

$$\text{Area on which the Earth is to be spread} = 120 - 20 = 100 \text{ m}^2$$

$$\text{Volume of Earth dug out} = 5 \times 4 \times 2 = 40 \text{ m}^3$$

$$\therefore \text{Level of field raised} = \frac{40}{100} = \frac{2}{5} \text{ m}$$

$$= \frac{2}{5} \times 100 = 40 \text{ cm}$$

38. (a) Edge of cube = 9 cm

For rectangular vessel,

Length = 15 cm and breadth = 12 cm

Let the rise in the water level is x cm.

Volume of cube = $9^3 = 9 \times 9 \times 9 \text{ cm}^3$

Volume of water risen in the vessel
 $= 15 \times 12 \times x \text{ cm}^3$

$$\Rightarrow 15 \times 12 \times x = 9 \times 9 \times 9$$

$$\therefore x = \frac{9 \times 9 \times 9}{15 \times 12} = \frac{81}{20} = 4.05 \text{ cm}$$

39. (b) The maximum diameter of a sphere in a cube is of 3 m.

$$\therefore \text{Volume of sphere, } V_1 = \frac{4}{3}\pi (1.5)^3$$

$$= 4.5\pi \text{ m}^3$$

$$\text{Volume of cube, } V_2 = (3)^3 = 27 \text{ m}^3$$

$$\text{and Volume of solid left} = V_2 - V_1 = (27 - 4.5\pi) \text{ m}^3$$

40. (c) Length of water flowing in 1h = 10 km

$$\therefore \text{Length of the water flowing in } \frac{1}{2} \text{ h}$$

$$= 5 \text{ km} = 5000 \text{ m}$$

Area of pipe

$$= 40 \text{ cm}^2 = \frac{40}{10000} \text{ m}^2 = \frac{1}{250} \text{ m}^2$$

$$\therefore \text{Volume of water flowing in } \frac{1}{2} \text{ h}$$

$$= 5000 \times \frac{1}{250} = 20 \text{ m}^3$$

$$\text{Area of bottom of tank} = 80 \times 40 = 3200 \text{ m}^2$$

$\therefore$ Depth of water

$$= \frac{\text{Volume}}{\text{Area}} = \frac{20}{3200} = \frac{1}{160} \text{ m}$$

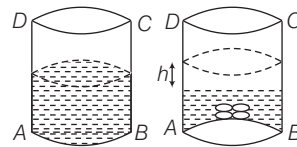
$$= \frac{1}{160} \times 100 \text{ cm} = \frac{5}{8} \text{ cm}$$

41. (c) Given radius of balls = 1 cm

$$\text{Volume of each ball} = \frac{4}{3} \times \pi \times 1 \times 1 \times 1 = \frac{4}{3}\pi \text{ cm}^3$$

$\therefore$ Volume of 4 balls

$$= 4 \times \frac{4}{3}\pi = \frac{16}{3}\pi \text{ cm}^3$$



Volume of water increased

= Volume of balls

$$\Rightarrow \text{Area of base} \times \text{height} = \frac{16}{3}\pi$$

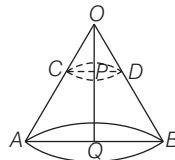
$$\Rightarrow \pi \times 5 \times 5 \times h = \frac{16}{3}\pi$$

$$\therefore h = \frac{16}{3 \times 25} = \frac{16}{75} \text{ cm}$$

42. (c) Given, height of the cone = 30 cm

$\therefore$ Volume of original cone (AOB)

$$= \frac{1}{3}\pi R^2 H$$



$$= \frac{1}{3} \times \pi R^2 \times 30 = 10\pi R^2 \text{ cm}^3$$

Volume of small cone OCD

$$= \frac{1}{27} \text{ of volume cone AOB}$$

$$\Rightarrow \frac{1}{3}\pi r^2 h = \frac{1}{27} \times 10\pi R^2$$

$$\Rightarrow h = \frac{10\pi R^2}{27} \times \frac{3}{\pi r^2} \Rightarrow h = \frac{10}{9} \left(\frac{R}{r}\right)^2$$

$$\therefore \frac{R}{r} = \frac{30}{h} \quad [\because \triangle OQB \sim \triangle OPD]$$

$$\therefore h = \frac{10}{9} \left(\frac{30}{h}\right)^2 = \frac{10}{9} \times \frac{900}{h^2}$$

$$\Rightarrow h^3 = 10 \times 100$$

$$\Rightarrow h = \sqrt[3]{1000} = 10 \text{ cm}$$

$\therefore$ Required height

$$= H - h = 30 - 10 = 20 \text{ cm}$$

43. (a) Let r be the radius of cylindrical block, then height will be $2r$.

$$\text{Volume of block} = \pi(r^2)(2r) = 2\pi r^3$$

A sphere of maximum possible volume is curved out whose radius will be r .

$$\therefore \text{Volume of sphere} = \frac{4}{3}\pi r^3$$

$$\therefore \text{Volume of utilised wood} = \frac{4}{3}\pi r^3$$

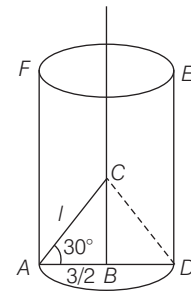
and volume of wasted wood

$$= 2\pi r^3 - \frac{4}{3}\pi r^3 = \frac{6\pi r^3 - 4\pi r^3}{3} = \frac{2\pi r^3}{3}$$

$$\therefore \text{Required ratio} = \frac{4}{3}\pi r^3 : \frac{2}{3}\pi r^3 = 2:1$$

44. (b) Let the slant height of cone = l cm

In $\triangle ABC$,



$$\cos 30^\circ = \frac{AB}{AC} = \frac{3/2}{l} \quad [\because AC = l \text{ cm}]$$

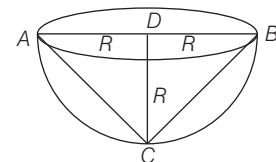
$$\Rightarrow l = \frac{3/2}{\sqrt{3}/2} = \sqrt{3} \text{ cm} \quad [\because \cos 30^\circ = \frac{\sqrt{3}}{2}]$$

$$\therefore \text{Area of cone } ACD = \pi r l$$

$$= \pi \times \frac{3}{2} \times \sqrt{3} = \frac{3\sqrt{3}\pi}{2} \text{ cm}^2$$

45. (a) Volume of cone, $C = \frac{1}{3}\pi R^2 H$

$$= \frac{1}{3}\pi R^3 \quad [\because H = R]$$



$$\text{Volume of hemisphere, } H = \frac{2}{3}\pi R^3$$

$$\therefore C : H = \frac{1}{3}\pi R^3 : \frac{2}{3}\pi R^3 = 1:2$$

46. (d) Volume of wire = $\pi r^2 h$

Now, new radius of the wire

$$= \frac{r \times 90}{100} = \frac{9r}{10}$$

Let new length of the wire be L_2 .

$$\therefore \text{Volume of new wire} = \pi \left(\frac{9r}{10}\right)^2 \times L_2$$

$$= \frac{81}{100}\pi r^2 L$$

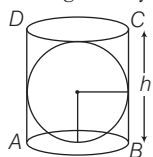
$$\text{By given condition, } \pi r^2 h = \frac{81}{100}\pi r^2 L$$

$$\Rightarrow L = \frac{100}{81} h$$

$$\text{Increases in length} = \frac{100}{81} b - b = \frac{19}{81} b$$

$$\text{Percentage increase} = \frac{19/81b}{b} \times 100\% = 23.45\% = 23\% \text{ (approx)}$$

47. (b) Let the height of cylinder be h .



$$\text{Then, radius of cylinder} = \frac{b}{2}$$

$$\text{Also, radius of the sphere, } r = \frac{b}{2}$$

$$\therefore \text{Volume of cylinder} = \pi \left(\frac{b}{2}\right)^2 h = \frac{\pi b^3}{4}$$

and volume of sphere.

$$S = \frac{4}{3} \pi r^3 = \frac{4}{3} \pi \left(\frac{b}{2}\right)^3 = \frac{\pi b^3}{6}$$

$$\therefore \text{Volume of remaining material} = \frac{\pi b^3}{4} - \frac{\pi b^3}{6} = \frac{\pi b^3}{12}$$

$\therefore$ Volume of sphere in remaining material,

$$S_1 = \frac{4}{3} \pi r_1^3$$

$$\text{By given condition, } \frac{4}{3} \pi r_1^3 = \frac{\pi b^3}{12}$$

$$\Rightarrow r_1^3 = \frac{b^3}{16} \Rightarrow r_1 = \frac{b}{2 \cdot 2^{1/3}}$$

$$\text{Required ratio} = \frac{r}{r_1} = \frac{b/2}{b/2 \cdot 2^{1/3}} = 2^{1/3}$$

$$\Rightarrow r : r_1 = 2^{1/3} : 1$$

48. (c) Let side of square be x

$\therefore$ Radius of sphere = x

Surface area of sphere, $A = 4\pi x^2$

Since, square revolves round a side to generate a cylinder whose height and radius are x and x , respectively.

$$\therefore S = 2\pi x(x+x) = 4\pi x^2$$

So, it is clear that, $A = S$.

49. (a) Given, length of pool = 24 m

and breadth of pool = 15 m

Rise in height of water

$$= 1 \text{ cm} = 0.01 \text{ m}$$

Volume of water displaced by x men

$$= 24 \times 15 \times 0.01 = 3.6 \text{ m}^3$$

But volume of water displaced by x men

$$= 0.1x \text{ m}^3$$

$$\therefore 0.1x = 3.6 \Rightarrow x = \frac{3.6}{0.1} = 36$$

50. (b) Given, length of one compartment = 200 m

and breadth of one compartment

$$= 100 \text{ m}$$

and height of water in one compartment

$$= 12 \text{ m}$$

Volume of one compartment

$$= 200 \times 100 \times 12$$

$$= 240000 \text{ m}^3$$

$\therefore$ Volume of 3 compartments

$$= 3 \times 240000 = 720000 \text{ m}^3$$

$$= 720000000 \text{ L}$$

$$[\because 1 \text{ m}^3 = 1000 \text{ L}]$$

Total requirement of water 50000

inhabitants in 1 day

$$= 50000 \times 20 = 1000000 \text{ L}$$

$\therefore$ Required number of days

$$= \frac{720000000}{1000000} = 720 \text{ days}$$

51. (b) Let r and h be the radius and height of the cone and r and H be the radius and height of the cylinder.

$$\therefore \text{Volume of cone} = \frac{1}{3} \pi r^2 h$$

and volume of cylinder = $\pi r^2 H$

$$\text{By given condition, } \frac{1}{3} \pi r^2 h = \pi r^2 H$$

$$\Rightarrow h = 3H \dots (i)$$

Lateral surface area of cone = $\pi r l$

and lateral surface area of cylinder

$$= 2\pi r H$$

$$\text{By given condition, } \pi r l = \frac{15}{8} \times 2\pi r H$$

$$\Rightarrow l = \frac{15}{4} H \Rightarrow l^2 = \frac{225}{16} H^2$$

$$\Rightarrow r^2 + b^2 = \frac{225}{16} H^2 [\because r^2 + b^2 = l^2]$$

$$\Rightarrow r^2 + 9H^2 = \frac{225}{16} H^2 \text{ [from Eq. (i)]}$$

$$\Rightarrow r^2 = \frac{81}{16} H^2 \Rightarrow \frac{r^2}{H^2} = \frac{81}{16}$$

$$\Rightarrow \frac{r}{H} = \frac{9}{4}$$

$$\therefore r : H = 9 : 4$$

52. (b) Let the sides of the cuboid be x , $2x$ and $4x$ and the side of the cube is y .

$$\therefore \text{Volume of cuboid} = x \times 2x \times 4x = 8x^3$$

and volume of cube = y^3

By given condition,

Volume of cuboid = Volume of cube

$$\therefore 8x^3 = y^3 \Rightarrow y = 2x \dots (i)$$

$\therefore$ Diagonal of cuboid

$$= \sqrt{x^2 + 4x^2 + 16x^2} = \sqrt{21}x$$

and diagonal of cube = $y\sqrt{3} = 2x\sqrt{3}$

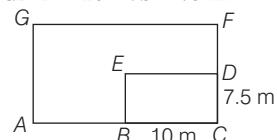
[from Eq. (i)]

Hence, the required ratio

$$= \frac{\sqrt{21}x}{2x\sqrt{3}} = \sqrt{\frac{21}{4 \times 3}} = \sqrt{1.75}$$

53. (c) Area of tank,

$$BCDE = 10 \times 7.5 = 75 \text{ m}^2$$



Area of remaining field

$$ABEDFGA = 125 \times 15 - 75 = 1800 \text{ m}^2$$

$$\text{Volume of Earth dug} = 10 \times 7.5 \times 6 = 450 \text{ m}^3$$

By given condition, $1800 \times b = 450$

$$\Rightarrow b = \frac{1}{4} \text{ m} = \frac{1}{4} \times 100 \text{ cm} = 25 \text{ cm}$$

54. (b) Let $r_1 = k$ and $r_2 = nk$

Since, $V_1 = V_2$

$$\therefore \frac{1}{3} \pi r_1^2 h_1 = \frac{1}{3} \pi r_2^2 h_2$$

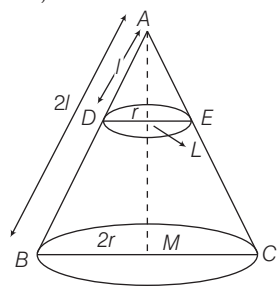
$$\Rightarrow \frac{1}{3} \pi k^2 \times h_1 = \frac{1}{3} \pi n^2 k^2 h_2$$

$$\Rightarrow h_1 = 3n^2 h_2$$

55. (d) Let r be the radius, h the height and l the slant height of smaller cone

$$\text{Given, } \frac{\text{slant height of original cone}}{\text{slant height of smaller cone}} = \frac{2}{1}$$

Since, $\triangle ABM$ and $\triangle ADL$ are similar.



Therefore, radius and height of original cone will be $2r$ and $2h$ respectively.

$$\therefore \frac{V_1}{V_2} = \frac{\frac{1}{3} \pi (2r)^2 \times 2h}{\frac{1}{3} \pi r^2 h} = \frac{8r^2 h}{r^2 h} = 8$$

$$\therefore V_1 : V_2 = 8 : 1$$

56. (d) By given condition,

Surface area of new sphere = 25% of 616

$$4\pi r^2 = 154$$

$$\Rightarrow r^2 = \frac{154}{22 \times 4} = \left(\frac{7}{2}\right)^2 \Rightarrow r = 3.5 \text{ cm}$$

57. (d) Let side of a cube be 'a' unit.

Then, radius of sphere is $\frac{a}{2}$ unit.

$$\therefore \frac{\text{Volume of cube}}{\text{Volume of sphere}} = \frac{(a)^3}{\frac{4\pi}{3} \left(\frac{a}{2}\right)^3} = \frac{6}{\pi}$$

58. (d) Total surface area = Curved surface area of cylinder + Curved surface area of cone + Top surface area of cylinder
 $= 2\pi rh + \pi rl + \pi r^2 = \pi [2rh + r^2 + rl]$
 $= \pi [2 \times 3 \times 4 + 3^2 + 3\sqrt{3^2 + 4^2}]$
 $= 48\pi \text{ cm}^2$

59. (b) Curved surface of a cylinder
 $= 1000 \text{ cm}^2, 2\pi rh = 1000 \dots(i)$

Length of wire used in one round

$$= \text{Perimeter of cylinder's base} = 2\pi r$$

$\therefore$ Number of rounds

$$= \frac{\text{Height of cylinder}}{\text{Diameter of wire}} = \frac{h}{0.5}$$

$\therefore$ Required length of wire

$$= 2\pi r \times \frac{h}{0.5} = \frac{2\pi rh}{0.5}$$

$$= \frac{1000}{0.5} = 2000 \text{ cm or } 20 \text{ m}$$

60. (d) Let h be the depth of the pit.

$\therefore$ Volume of Earth dug

$$= 500 \times 250 \text{ cm}^3 = 125000 \text{ cm}^3$$

But volume of pit = $50 \times 50 \times h$

$$\therefore h = \frac{125000}{50 \times 50} = 50 \text{ cm} = 0.5 \text{ m}$$

61. (a) Since, sheet is revolved about its length, therefore a cylinder is formed with
 $h = 7 \text{ cm}$ and $r = 4 \text{ cm}$

$\therefore$ Volume of the figure, thus formed

$$= \pi r^2 h = \frac{22}{7} \times 4 \times 4 \times 7 = 352 \text{ cm}^3$$

62. (c) Let l, b and h be the sides of cuboid.

then, $l^2 + b^2 = x^2 \dots(i)$

$$b^2 + h^2 = y^2 \dots(ii)$$

and $h^2 + l^2 = z^2 \dots(iii)$

Adding Eq. (i), (ii) and (iii), we get

$$2(l^2 + b^2 + h^2) = x^2 + y^2 + z^2$$

$$\Rightarrow l^2 + b^2 + h^2 = \frac{1}{2}(x^2 + y^2 + z^2)$$

$\dots(iv)$

From Eqs. (i), (ii), (iii) and (iv)

$$h = \sqrt{\frac{y^2 + z^2 - x^2}{2}}, l = \sqrt{\frac{z^2 + x^2 - y^2}{2}}$$

and

$$b = \sqrt{\frac{x^2 + y^2 - z^2}{2}}$$

Hence, volume of cuboid = lbh

$$= \sqrt{\frac{(y^2 + z^2 - x^2)(z^2 + x^2 - y^2)(x^2 + y^2 - z^2)}{2 \times 2 \times 2}}$$

$$= \frac{1}{2\sqrt{2}} \sqrt{(y^2 + z^2 - x^2)(z^2 + x^2 - y^2)(x^2 + y^2 - z^2)}$$

63. (a) Let the radius of cone and sphere be r and height of cone be h_1 .

By given condition,

Volume of cone = Volume of sphere

$$\therefore \frac{1}{3} \pi r^2 h_1 = \frac{4}{3} \pi (r)^3 \Rightarrow \frac{h_1}{2r} = \frac{2}{1}$$

or, $2 : 1$

64. (d) Let the length, breadth and height of a rectangular parallelopiped be $6x, 3x$ and x , respectively.

Also, let the side of a cube be a .

By given condition,

Surface area of a cube = Surface area of rectangular parallelopiped

$$\Rightarrow 6(a)^2 = 2(6x \times 3x + 3x \times x + x \times 6x)$$

$$\Rightarrow 6a^2 = 2(18x^2 + 3x^2 + 6x^2)$$

$$\Rightarrow 6a^2 = 54x^2$$

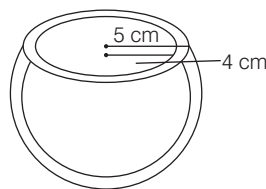
$$\therefore a = 3x$$

Now,

$$\frac{\text{Volume of cube}}{\text{Volume of rectangular parallelopiped}} = \frac{a^3}{6x \times 3x \times x} = \frac{(3x)^3}{18x^3} = \frac{27x^3}{18x^3} = \frac{3}{2}$$

or, $3 : 2$

65. (b) Given thickness of a metal = 1 cm



$\therefore$ Inner radius

= outer radius - thickness

$$= 5 - 1 = 4 \text{ cm}$$

$$\text{Volume of hemisphere} = \frac{2}{3} \pi (R^3 - r^3)$$

$$= \frac{2}{3} \pi (5^3 - 4^3) = \frac{2}{3} \pi (125 - 64)$$

$$= \frac{2}{3} \pi \times 61 \text{ cm}^3$$

$\therefore$ Weight of 1 cm^3 of metal = 9 g

$\therefore$ Weight of all metal

$$= \frac{2}{3} \pi \times 61 \times 9 \text{ g} = 366 \pi \text{ g}$$

66. (b) Let the radius of ball be r .

$\therefore$ Radius of base of cylinder = $4r$

and height of cylinder = $4r$

$$\therefore \text{Volume of spherical ball} = \frac{4}{3} \pi r^3$$

and volume of water

$$= \frac{1}{2} \pi (4r)^2 (4r) = 32 \pi r^3$$

Also, volume of remaining portion of cylinder = $32 \pi r^3$

Let number of spherical balls be n .

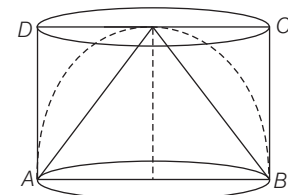
$$\therefore 32 \pi r^3 = n \times \frac{4}{3} \pi r^3$$

$$\Rightarrow n = 8 \times 3 = 24$$

67. (b) From the information given in the question and the figure it is clear that

Radius of the hemisphere = radius of cone = height of cone = height of cylinder. Let it be r .

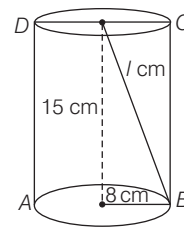
Then, ratio of volume of cylinder, hemisphere and cone.



$$= \pi r^3 : \frac{2}{3} \pi r^3 : \frac{1}{3} \pi r^3$$

$$= 1 : \frac{2}{3} : \frac{1}{3} = 3 : 2 : 1$$

68. (a) $l = \sqrt{15^2 + 8^2} = 17 \text{ cm}$



Total surface area of the remaining solid

$$= 2\pi rh + \pi r^2 + \pi rl$$

$$= 2\pi \times 8 \times 15 + \pi(8)^2 + \pi \times 8 \times 17$$

$$= 240\pi + 64\pi + 136\pi$$

$$= 440 \pi \text{ cm}^2$$

69. (c) Volume of spherical lead shot

$$= \frac{4}{3} \pi (1)^3 = \frac{4}{3} \pi \text{ cm}^3$$

I. Volume of 8 shots

$$= \frac{4}{3} \pi (0.5)^3 \times 8$$

$$= \frac{4}{3} \pi \text{ cm}^3$$